Data Flow Diagram & User Stories

Requirement Analysis Phase — Phase II

Project Name	Automated Network Request Management in ServiceNow
Phase	Phase 2: Requirement Analysis
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Date	29 June 2026
Platform / Domain	ServiceNow / IT Service Management (ITSM)
Academic Year	2025 – 2026

1. Data Flow Diagram (DFD) — Structural Map

The following structural map illustrates how data enters, moves through, and leaves the Automated Network Request Management system. Since a DFD is fundamentally a visual model, this textual representation captures all external entities, data flows, data stores, and processing nodes.

External Entities

- Requester (IT Staff / End-User)
- Approver (Line Manager / Security Team Lead)
- Fulfillment Team (Network Engineer)

Data Flows

Inputs: Request parameters — IP addresses, port numbers, VPN profile selections, VLAN identifiers, business justifications.

Outputs: Approval notification emails, assigned catalog tasks (sc_task), completion alerts, and status update payloads.

Data Stores

- sc_req_item — Requested Item Table (stores all catalog request submissions).
- sysapproval_approver — Approval Table (stores all approval decisions and comments).
- sc_task — Catalog Task Table (stores all fulfillment tasks assigned to engineers).
- u_network_database — Custom logging table (tracks request numbers, device types, work status, and update counters).

2. DFD Flow Summary

DFD Level / Process	Flow Description
Level 0 (Context)	[Requester] → Submit Form Data → [System] → Notifications/Tasks → [Approver / Engineer]
Level 1: Process 1.0	[Requester] → Submits Form Data → [1.0 Process Intake] ↔ [sc_req_item]
Level 1: Process 2.0	[1.0] → Triggers Flow Engine → [2.0 Process Approvals] ↔ [sysapproval_approver] ← [Approver]
Level 1: Process 3.0	[2.0 Approved] → [3.0 Generate Task] ↔ [sc_task] → [Network Engineer]
Level 1: Process 4.0	[3.0 Task Complete] → [4.0 Close Request] → [Requester Notification] + [Audit Log]

3. User Stories

US-001 — IT Staff / Requester

User Story: As an IT staff member, I want to submit a structured network access request through the Service Catalog so that my request is captured completely and routed automatically without manual email chains.

Acceptance Criteria:

- Service Catalog item 'Network Access Request' is accessible from the Service Portal.
- Form captures: Requester name, department, target resource, business justification, required date.
- Submission triggers an immediate confirmation email with a REQ reference number.
- Request record is created in sc_req_item and Flow Designer workflow is triggered.

US-002 — Line Manager / Approver

User Story: As a line manager, I want to receive a structured email notification with full request context so that I can approve or reject network requests quickly from my inbox or the ServiceNow workspace.

Acceptance Criteria:

- Approval email contains: requester name, request type, business justification, and direct Approve/Reject action links.
- Approval decision is recorded in sysapproval_approver with timestamp and manager comments.
- Requester is notified automatically upon manager decision.
- If rejected, requester receives reason and optional resubmission guidance.

US-003 — Network Security Team

User Story: As a network security reviewer, I want to receive high-risk requests (firewall changes, VLAN modifications) for technical vetting after manager approval so that security governance is enforced before fulfillment.

Acceptance Criteria:

- Flow Designer applies conditional routing to escalate high-risk items to the Security Team queue.
- Security team receives structured email notification with full technical parameters.
- Security approval or rejection is recorded with mandatory comments.
- Upon security approval, fulfillment task is auto-generated in sc_task.

US-004 — Network Engineer

User Story: As a network engineer, I want to receive a clearly defined fulfillment task with all technical parameters pre-populated so that I can execute the required network change without seeking clarification.

Acceptance Criteria:

- sc_task record contains: request type, IP/VLAN/port details, business justification, SLA deadline, and requester contact.

- Task is automatically assigned to the correct Network Operations Group queue via Assignment Rules.
- Engineer updates the task with completion notes upon implementation.
- Task closure triggers automatic closure of the parent sc_req_item and dispatches completion notification to the requester.

US-005 — IT Administrator

User Story: As an IT administrator, I want to monitor all network request records and SLA performance through dashboards so that I can ensure compliance, identify bottlenecks, and generate management reports.

Acceptance Criteria:

- Admin dashboard displays: open request count, pending approvals, active tasks, and SLA breach alerts.
- All request records are fully auditable with immutable timestamps on every state change.
- Reports exportable for compliance and management review.
- SLA escalation emails sent automatically to team leads at 50%, 75%, and 100% breach thresholds.